



American International University-Bangladesh (AIUB)
Department of Computer Science Faculty of Science & Technology (FST)

FIRMLINK

A Software Engineering Project Submitted
By

Semester-->Summer-25-26		Section:	
SL	Student Name	Student ID	
1	Tonmoy chakrabarty joy	24-57838-2	
2	Samia mahmood	22-49646-3	
3	Prottoy mirza	23-50154-1	
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1. PROJECT PROPOSAL

1.1 Background to the Problem

In the contemporary agricultural infrastructure of Bangladesh, traditional supply chains create an artificial, highly inefficient barrier between rural smallholder farmers and their target urban consumer base. Despite investing intensive physical labor and capital to harvest high-yield, premium-quality crops, local farmers continue to operate under severe socio-economic vulnerabilities driven by deep-rooted market exploitation and operational isolation.

The core structural failure within this domain is the multi-tiered network of traditional intermediaries and middlemen, locally known as Farias, Beparis, and Aratdars. These unregulated brokers intercept the supply chain, purchasing commodities at unsustainably low rates directly at the farmgate and systematically inflating consumer retail prices by up to 60%. As a consequence, actual producers remain trapped in cyclical poverty with depleted profit margins, while urban buyers face extreme, artificial food inflation.

This system is further broken by widespread information asymmetry. Rural farmers lack access to localized, real-time market data or baseline consumer rates, making it impossible to reject unfair offers. On the logistics front, a lack of automated, on-demand shipping systems leaves micro-producers physically isolated from major commercial hubs, leading to significant post-harvest losses and waste before goods ever reach the city markets.

1.2 Target Group of Users

Rural Farmers: Smallholder and multi-crop agricultural producers who require direct, unmediated access to open markets to list seasonal yields, establish independent pricing authority, and maximize baseline revenues without intermediary interference.

Urban Buyers: Individual retail consumers, localized neighborhood grocery entities, and commercial B2B hospitality businesses seeking transparent, highly fresh, and fully traceable farm produce delivered at competitive, non-inflated rates.

Commission-Based Local Agents: Tech-literate rural youth and local tech-enablers onboarded to serve as operational proxies for digitally disconnected or older farmers, managing virtual catalogs, stock inventory entries, and high-volume wholesale negotiations.